

Local Ancestry using Relate

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1 Log-likelihood

$$\log P(t, z, q | \gamma) = \sum_{l=1}^T \sum_{i=1}^{\#coal(l)} \sum_{j=1}^A \sum_{k=1}^{N_{li}} \sum_{m=1}^E I_{\{q_{li}=k\}} I_{\{z_l=j\}} I_{\{e_{li}=m\}} \log P(z_l = j, q_{li} = k, t_i | \gamma) \quad (1)$$

where,

T = total number of trees,

$\#coal(l)$ = number of coalescence events with the target in l^{th} tree,

A = number of clusters assumed in the target,

N_{li} = number of reference individuals in i^{th} coalescence event,

E = number of epochs,

z_l = local ancestry of the target at position l ,

q_{li} = reference individual index that coalesces in the i^{th} coalescence event,

e_{li} = epoch in which i^{th} coalescent event happens in the l^{th} tree.

Simplifying the log-likelihood further as:

$$P(z_l = j, q_{li} = k, t_i | \gamma) = P(t_i | \gamma, z_l = j, q_{li} = k) P(q_{li} = k | z_l = j, \gamma) P(z_l = j | \gamma) \quad (2)$$

Looking at the first term $P(t_i | \gamma, z_l = j, q_{li} = k)$:

$$P(t_i | \gamma, z_l = j, q_{li} = k) = P(t_i | \gamma, z_l = j) \quad (3)$$

$$P(t_i | \gamma, z_l = j) = \frac{\sum_{g=1}^R \gamma_{jg}(m) n_{ig}}{\sum_{g=1}^R n_{ig}} e^{-\sum_{g=1}^R \gamma_{jg} \cdot O_{lg}[i]} \quad (4)$$

where,

γ_{jg} = coalescence rate between j^{th} cluster and g^{th} reference group,

n_{ig} = effective ¹ number of group g individuals coalescing in the i^{th} coalescence event,

$O_{lg}[i]$ = sum of branch length corresponding to group g individuals in the i^{th} coalescence event.

¹weigh each lineage by its past coalescence history with other lineages

The second term $P(q_{li} = k | z_l = j, \gamma)$ simplifies to:

$$P(q_{li} = k | z_l = j, \gamma) = \frac{\gamma_{jg(k)}(m) C_{ik} \sum_{g=1}^R n_{ig}}{\sum_{g=1}^R \gamma_{jg}(m) n_{ig}} \quad (5)$$

where, $g(k)$ means the group assignment (among R groups) of the k^{th} individual in the reference panel, and C_{ik} is the contribution of k^{th} lineage in i^{th} clade. The third term $P(z_l = j | \gamma)$ is independent of γ and written as follows:

$$P(z_l = j | \gamma) = \pi_j \quad (6)$$

The overall likelihood looks like:

$$P(z_l = j, q_{li} = k, t_i | \gamma) = \frac{\pi_j \gamma_{jg(k)}(m) C_{ik}}{\sum_{g=1}^R n_{ig}} e^{-\sum_{g=1}^R \gamma_{jg} \cdot O_{lg}[i]} \quad (7)$$

$$\propto \gamma_{jg(k)}(m) * e^{-\sum_{g=1}^R \gamma_{jg} \cdot O_{lg}[i]} \quad (8)$$

2 E-step

$$E_{z,q|\gamma^{t-1},t} \left(\sum_{l=1}^T \sum_{i=1}^{\#coal(l)} \sum_{j=1}^A \sum_{k=1}^{N_{li}} \sum_{m=1}^E I_{\{q_{li}=k\}} I_{\{z_l=j\}} I_{\{e_{li}=m\}} \log P(z_l = j, q_{li} = k, t_i|\gamma) \right) \quad (9)$$

$$= \sum_{l=1}^T \sum_{i=1}^{\#coal(l)} \sum_{j=1}^A \sum_{k=1}^{N_{li}} \sum_{m=1}^E E_{z_l, q_{li}|\gamma^{t-1},t} (I_{\{q_{li}=k\}} I_{\{z_l=j\}} I_{\{e_{li}=m\}} \log(P(z_l = j, q_{li} = k, t_i|\gamma))) \quad (10)$$

$$= \sum_{l=1}^T \sum_{i=1}^{\#coal(l)} \sum_{j=1}^A \sum_{k=1}^{N_{li}} \sum_{m=1}^E P(q_{li} = k, z_l = j|\gamma^{t-1}, t) I_{\{e_{li}=m\}} \log(P(z_l = j, q_{li} = k, t_i|\gamma)) \quad (11)$$

$$= \sum_{l=1}^T \sum_{i=1}^{\#coal(l)} \sum_{j=1}^A \sum_{k=1}^{N_{li}} \sum_{m=1}^E \left(P(q_{li} = k|\gamma^{t-1}, t, z_l = j) P(z_l = j|\gamma^{t-1}) I_{\{e_{li}=m\}} \log(P(z_l = j, q_{li} = k, t_i|\gamma)) \right) \quad (12)$$

$$= \sum_{l=1}^T \sum_{i=1}^{\#coal(l)} \sum_{j=1}^A \sum_{k=1}^{N_{li}} \sum_{m=1}^E \left(P(q_{li} = k|\gamma^{t-1}, z_l = j) P(z_l = j|t, \gamma^{t-1}) \right.$$

$$\left. I_{\{e_{li}=m\}} \left(\log(\gamma_{jg(k)}(m)) - \sum_{g=1}^R \gamma_{jg} \cdot O_{lg}[i] \right) \right) \quad (13)$$

$$= \sum_{l=1}^T \sum_{i=1}^{\#coal(l)} \sum_{j=1}^A \sum_{k=1}^{N_{li}} \sum_{m=1}^E \left(P(q_{li} = k|\gamma^{t-1}, z_l = j) P(z_l = j|t, \gamma^{t-1}) I_{\{e_{li}=m\}} \log(\gamma_{jg(k)}(m)) \right) \\ - \sum_{l=1}^T \sum_{i=1}^{\#coal(l)} \sum_{j=1}^A \sum_{k=1}^{N_{li}} \sum_{m=1}^E \left(P(q_{li} = k|\gamma^{t-1}, z_l = j) P(z_l = j|t, \gamma^{t-1}) I_{\{e_{li}=m\}} \sum_{g=1}^R \gamma_{jg} \cdot O_{lg}[i] \right) \quad (14)$$

$$= \sum_{l=1}^T \sum_{i=1}^{\#coal(l)} \sum_{j=1}^A \sum_{g=1}^R \sum_{m=1}^E \left(P(q'_{li} = g|\gamma^{t-1}, z_l = j) P(z_l = j|t, \gamma^{t-1}) I_{\{e_{li}=m\}} \log(\gamma_{jg}(m)) \right) \\ - \sum_{l=1}^T \sum_{i=1}^{\#coal(l)} \sum_{j=1}^A \sum_{m=1}^E \left(P(z_l = j|t, \gamma^{t-1}) I_{\{e_{li}=m\}} \sum_{g=1}^R \gamma_{jg} \cdot O_{lg}[i] \right) \quad (15)$$

We group the individuals based on group assignment in the first term and use $\sum_{k=1}^{N_{li}} P(q_{li} = k|\gamma^{t-1}, z_l = j) = 1$ in the second term. Lets denote $U_{l,i,g,j}^{t-1}$ as the first term corresponding to the membership "group g individual coalesces in the i^{th} coalescence event". And, lets denote $T_{l,j}^{t-1}$ as the second term corresponding to the local ancestry of the target.

We use Bayes in order to evaluate $T_{l,j}$ given the previous γ as:

$$T_{l,j}^{t-1} = P(z_l = j | \gamma^{t-1}, t) = \frac{\prod_{i=1}^{\#coal(l)} P(t_i | z_l = j, \gamma^{t-1}) P(z_l = j | \gamma^{t-1})}{\sum_{j=1}^A \prod_{i=1}^{\#coal(l)} P(t_i | z_l = j, \gamma^{t-1}) P(z_l = j | \gamma^{t-1})} \quad (16)$$

$$= \frac{\pi_j^{t-1} \prod_{i=1}^{\#coal(l)} (\sum_{g=1}^R \gamma_{jg}^{t-1} n_{ig}) e^{-\sum_{g=1}^R \gamma_{jg}^{t-1} \cdot O_{lg}[i]}}{\sum_{j=1}^A \pi_j^{t-1} \prod_{i=1}^{\#coal(l)} (\sum_{g=1}^R \gamma_{jg}^{t-1} n_{ig}) e^{-\sum_{g=1}^R \gamma_{jg}^{t-1} \cdot O_{lg}[i]}} \quad (17)$$

$$= \frac{\left(\prod_{i=1}^{\#coal(l)} (\sum_{g=1}^R \gamma_{jg}^{t-1} n_{ig}) \right) \pi_j^{t-1} e^{-\sum_{g=1}^R \gamma_{jg}^{t-1} \cdot Opp_{lg}}}{\sum_{j=1}^A \left(\prod_{i=1}^{\#coal(l)} (\sum_{g=1}^R \gamma_{jg}^{t-1} n_{ig}) \right) \pi_j^{t-1} e^{-\sum_{g=1}^R \gamma_{jg}^{t-1} \cdot Opp_{lg}}} \quad (18)$$

where, Opp_{lg} is the overall opportunity matrix for group g in the l^{th} tree ². Evaluating $U_{l,i,g,j}^{t-1}$ now:

$$U_{l,i,g,j}^{t-1} = P(q'_{li} = g | \gamma^{t-1}, z_l = j) = \frac{\gamma_{jg}^{t-1}(m) n_{ig}}{\sum_{g=1}^R \gamma_{jg}^{t-1}(m) n_{ig}} \quad (19)$$

²It is E dimensional vector, whose i^{th} entry records total branch length from 1 to epoch i

3 M-step

Starting with the expected log-likelihood constructed in the E-step equation 12:

$$\sum_{l=1}^T \sum_{i=1}^{\#coal(l)} \sum_{j=1}^A \sum_{k=1}^{N_{li}} \sum_{m=1}^E \left(P(q_{li} = k | \gamma^{t-1}, t, z_l = j) P(z_l = j | t, \gamma^{t-1}) I_{\{e_{li}=m\}} \log(P(z_l = j, q_{li} = k, t_i | \gamma)) \right) \quad (20)$$

$$\begin{aligned} &= \sum_{l=1}^T \sum_{i=1}^{\#coal(l)} \sum_{j=1}^A \sum_{g=1}^R \sum_{m=1}^E \left(P(q'_{li} = g | \gamma^{t-1}, z_l = j) P(z_l = j | t, \gamma^{t-1}) I_{\{e_{li}=m\}} \log(\gamma_{jg}(m)) \right) - \\ &\quad \sum_{l=1}^T \sum_{i=1}^{\#coal(l)} \sum_{j=1}^A \sum_{m=1}^E \left(P(z_l = j | t, \gamma^{t-1}) I_{\{e_{li}=m\}} \sum_{g=1}^R \gamma_{jg} \cdot O_{lg}[i] \right) \quad (21) \\ &= \sum_{l=1}^T \sum_{i=1}^{\#coal(l)} \sum_{j=1}^A \sum_{g=1}^R \sum_{m=1}^E \left(U_{l,i,g,j}^{t-1} T_{l,j} I_{\{e_{li}=m\}} \log(\gamma_{jg}(m)) \right) - \sum_{l=1}^T \sum_{i=1}^{\#coal(l)} \sum_{j=1}^A \sum_{m=1}^E \left(T_{l,j} I_{\{e_{li}=m\}} \sum_{g=1}^R \gamma_{jg} \cdot O_{lg}[i] \right) \quad (22) \end{aligned}$$

Maximizing w.r.t $\gamma_{jg}(m)$ we get:

$$\sum_{l=1}^T \sum_{i=1}^{\#coal(l)} \left(\frac{I_{\{e_{li}=m\}} U_{l,i,g,j}^{t-1} T_{l,j}}{\gamma_{jg}(m)} \right) - \sum_{l=1}^T \sum_{i=1}^{\#coal(l)} \left(I_{\{e_{li}=m\}} T_{l,j} O_{lg}[i] \right) = 0 \quad (23)$$

$$\sum_{l=1}^T \sum_{i=1}^{\#coal(l)} \left(\frac{I_{\{e_{li}=m\}} U_{l,i,g,j}^{t-1} T_{l,j}}{\gamma_{jg}(m)} \right) = \sum_{l=1}^T \sum_{i=1}^{\#coal(l)} \left(I_{\{e_{li}=m\}} T_{l,j} O_{lg}[i] \right) \quad (24)$$

$$\gamma_{jg}^t(m) = \frac{\sum_{l=1}^T \sum_{i=1}^{\#coal(l)} I_{\{e_{li}=m\}} U_{l,i,g,j}^{t-1} T_{l,j}}{\sum_{l=1}^T \sum_{i=1}^{\#coal(l)} I_{\{e_{li}=m\}} T_{l,j} O_{lg}[i]} \quad (25)$$

$$\gamma_{jg}^t(m) = \frac{\sum_{l=1}^T \sum_{i=1}^{\#coal(l)} I_{\{e_{li}=m\}} U_{l,i,g,j}^{t-1} T_{l,j}}{\sum_{l=1}^T T_{l,j} Opp_{lg}(m)} \quad (26)$$

$$\gamma_{jg}^t(m) = \frac{\sum_{l=1}^T \sum_{i=1}^{\#coal(l)} I_{\{e_{li}=m\}} \frac{\gamma_{jg}^{t-1}(m)^{n_{ig}}}{\sum_{g=1}^R \gamma_{jg}^{t-1}(m)^{n_{ig}}} T_{l,j}}{\sum_{l=1}^T T_{l,j} Opp_{lg}(m)} \quad (27)$$

where, $Opp_{lg}(m)$ is the opportunity matrix in epoch m . We don't need to store $U_{l,i,g,j}^{t-1}$ explicitly, we can instead store γ^{t-1} to save on memory.